3

RS-485 fundamentals

Objectives

When you have completed study of this chapter, you will be able to:

- Describe the RS-485 standard
- Remedy the following problems:
- Incorrect RS-485 wiring
- Excessive common mode voltage
- Faulty converters
- Isolation
- Idle state problems
- Incorrect or missing terminations
- RTS control via hardware or software

3.1 The RS-485 interface standard

The RS-485-A standard is one of the most versatile of the RS interface standards. It is an extension of RS-422 and allows the same distance and data speed but increases the number of transmitters and receivers permitted on the line. RS-485 permits a 'multidrop' network connection on 2 wires and allows reliable serial data communication for:

- Distances of up to 1200 m (4000 feet, same as RS-422)
- Data rates of up to 10 Mbps (same as RS-422)
- Up to 32 line drivers on the same line
- Up to 32 line receivers on the same line

The maximum bit rate and maximum length can, however, not be achieved at the same time. For 24 AWG twisted pair cable the maximum data rate at 4000 ft (1200 m) is approximately 90 kbps. The maximum cable length at 10 Mbps is less than 20 ft (6m). Better performance will require a higher-grade cable and possibly the use of active (solid state) terminators in the place of the 120-ohm resistors.

According to the RS-485 standard, there can be 32 'standard' transceivers on the network. Some manufacturers supply devices that are equivalent to ½ or ¼ standard

device, in which case this number can be increased to 64 or 128. If more transceivers are required, repeaters have to be used to extend the network.

The two conductors making up the bus are referred to as A in B in the specification. The A conductor is alternatively known as A–, TxA and Tx+. The B conductor, in similar fashion, is called B+, TxB and Tx–. Although this is rather confusing, identifying the A and B wires are not difficult. In the MARK or OFF state (i.e. when the RS-232 TxD pin is LOW (e.g. minus 8 V), the voltage on the A wire is more negative than that on the B wire.

The differential voltages on the A and B outputs of the driver (transmitter) are similar (although not identical) to those for RS-422, namely:

- –1.5V to –6V on the A terminal with respect to the B terminal for a binary 1 (MARK or OFF) state
- +1.5V to +6V on the A terminal with respect to the B terminal for a binary 0 (SPACE or ON state)

As with RS-422, the line driver for the RS-485 interface produces a ±5V differential voltage on two wires.

The major enhancement of RS-485 is that a line driver can operate in three states called tri-state operation:

- Logic 1
- Logic 0
- High-impedance

In the high impedance state, the line driver draws virtually no current and appears not to be present on the line. This is known as the 'disabled' state and can be initiated by a signal on a control pin on the line driver integrated circuit. Tri-state operation allows a multidrop network connection and up to 32 transmitters can be connected on the same line, although only one can be active at any one time. Each terminal in a multidrop system must be allocated a unique address to avoid conflicting with other devices on the system. RS-485 includes current limiting in cases where contention occurs.

The RS-485 interface standard is very useful for systems where several instruments or controllers may be connected on the same line. Special care must be taken with the software to coordinate which devices on the network can become active. In most cases, a master terminal, such as a PC or computer, controls which transmitter/receiver will be active at a given time.

The two-wire data transmission line does not require special termination if the signal transmission time from one end of the line to the other end (at approximately 200 meters per microsecond) is significantly smaller than one quarter of the signal's rise time. This is typical with short lines or low bit rates. At high bit rates or in the case of long lines, proper termination becomes critical. The value of the terminating resistors (one at each end) should be equal to the characteristic impedance of the cable. This is typically 120 Ohms for twisted pair wire.

Figure 3.1 shows a typical two-wire multidrop network. Note that the transmission line is terminated on both ends of the line but not at drop points in the middle of the line.

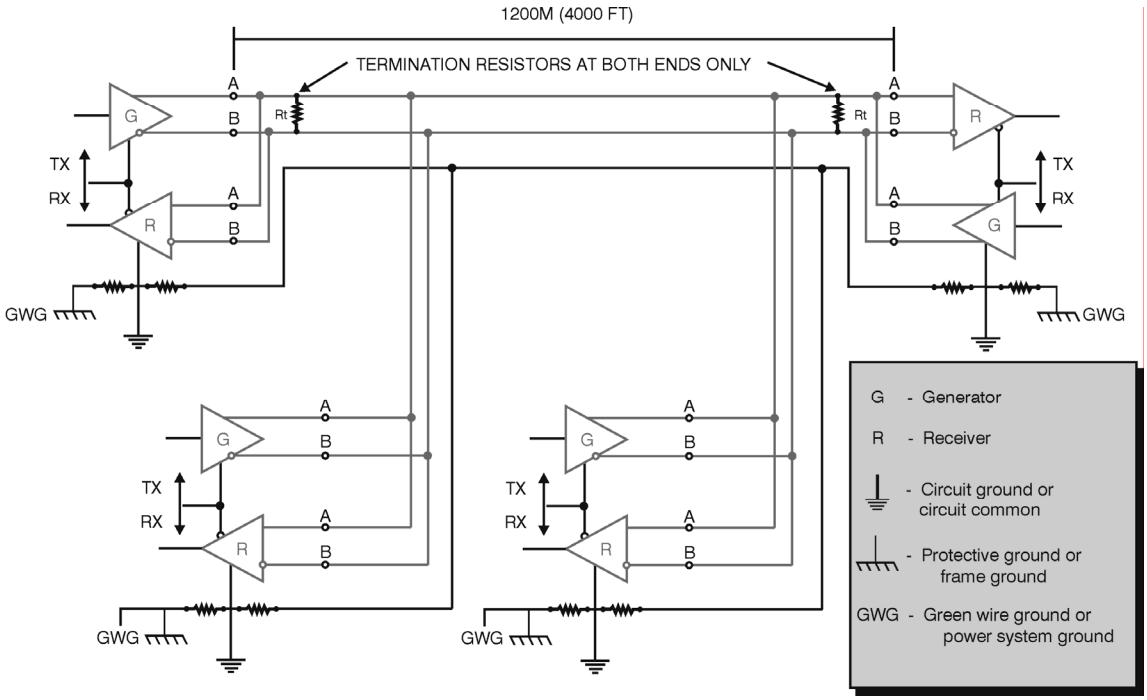


Figure 3.1
Typical two wire multidrop network

An RS-485 network can also be connected in a four wire configuration as shown in figure 3.2. In this type of connection it is necessary that one node is a master node and all others slaves. The master node communicates to all slaves, but a slave node can communicate only to the master. Since the slave nodes never listen to another slave's response to the master, a slave node cannot reply incorrectly to another slave node. This is an advantage in a mixed protocol environment.

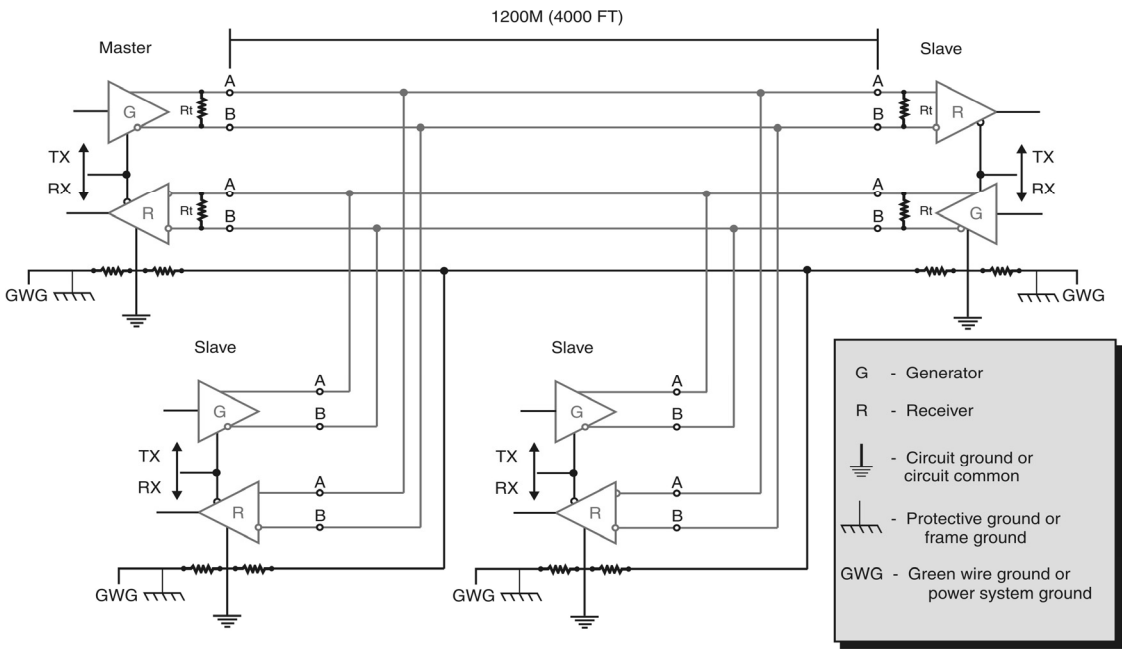


Figure 3.2
Four wire network configuration

During normal operation there are periods when all RS-485 drivers are off, and the communications lines are in the idle, high impedance state. In this condition the lines are susceptible to noise pick up, which can be interpreted as random characters on the communications line. If a specific RS-485 system has this problem, it should incorporate bias resistors, as indicated in figure 3.3. The purpose of the bias resistors is not only to reduce the amount of noise picked up, but to keep the receiver biased in the IDLE state when no input signal is received. For this purpose the voltage drop across the 120 Ohm termination resistor must exceed 200 mV AND the A terminal must be more negative than the B terminal. Keeping in mind that the two 120-Ohm resistors appear in parallel, the bias resistor values can be calculated using Ohm's Law. For a +5V supply and 120-Ohm terminators, a bias resistor value of 560 Ohm is sufficient. This assumes that the bias resistors are only installed on ONE node.

Some commercial systems use higher values for the bias resistors, but then assume that all or several nodes have bias resistors attached. In this case the value of all the bias resistors in parallel must be small enough to ensure 200 mV across the A and B wires.

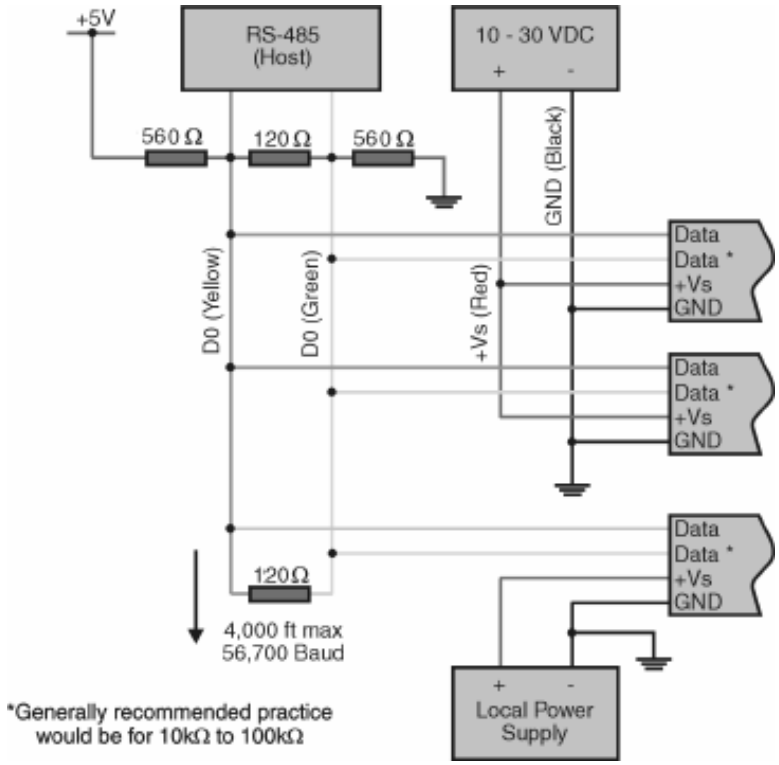


Figure 3.3
Suggested installation of resistors to minimize noise

RS-485 line drivers are designed to handle 32 nodes. This limitation can be overcome by employing an RS-485 repeater connected to the network. When data occurs on either side of the repeater, it is transmitted to the other side. The RS-485 repeater transmits at full voltage levels, consequently another 31 nodes can be connected to the network. A diagram for the use of RS-485 with a bi-directional repeater is given in figure 3.4.

The 'gnd' pin of the RS-485 transceiver should be connected to the logic reference (also known as circuit ground or circuit common), either directly or through a 100-Ohm ½ Watt resistor. The purpose if the resistor is to limit the current flow if there is a significant potential difference between the earth points. This is not shown in figure 3.2. In addition, the logic reference is to be connected to the chassis reference (protective

ground or frame ground) through a 100 Ohm 1/2 Watt resistor. The chassis reference, in turn, is connected directly to the safety reference (green wire ground or power system ground).

If the grounds of the nodes are properly interconnected, then a third wire running in parallel with the A and B wires are technically speaking not necessary. However, this is often not the case and thus a third wire is added as in figure 3.2. If the third wire is added, a 100-ohm ½ W resistor is to be added at each end as shown in figure 3.2.

The 'drops' or 'spurs' that interconnect the intermediate nodes to the bus need to be as short as possible since a long spur creates an impedance mismatch, which leads to unwanted reflections. The amount of reflection that can be tolerated depends on the bit rate. At 50 kbps a spur of, say, 30 meters could be in order, whilst at 10 Mbps the spur might be limited to 30 cm. Generally speaking, spurs on a transmission line are "bad news" because of the impedance mismatch (and hence the reflections) they create, and should be kept as short as possible.

Some systems employ RS-485 in a so-called 'star' configuration. This is not really a star, since a star topology requires a hub device at its center. The 'star' is in fact a very short bus with extremely long spurs, and is prone to reflections. It can therefore only be used at low bit rates.

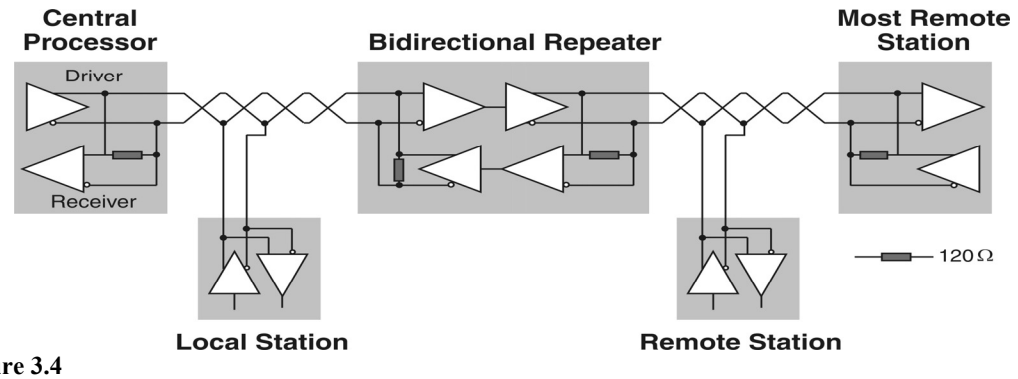


Figure 3.4
RS-485 used with repeaters

The 'decision threshold' of the RS-485 receiver is identical to that of both RS-422 & RS-423 receivers (not discussed as they have been superseded by RS-423) at 400 mV (0.4 V), as indicated in figure 3.5.

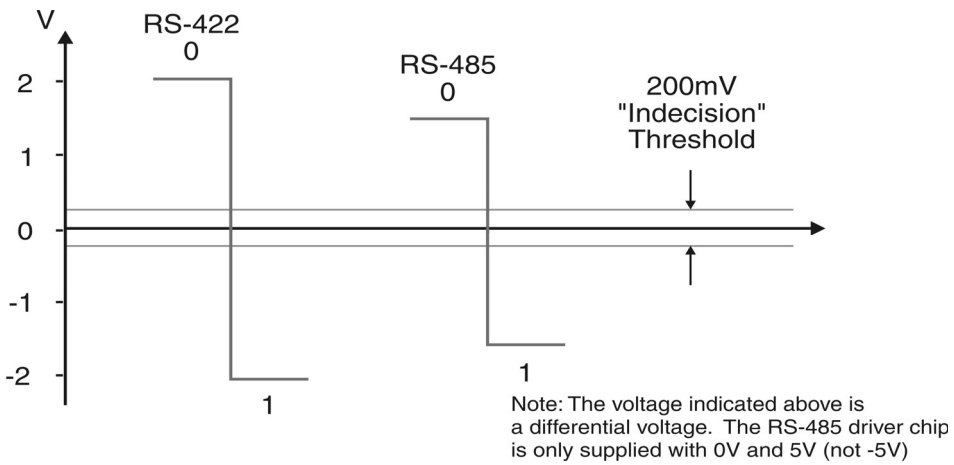


Figure 3.5
RS-485/422 & 423 receiver sensitivities

3.2 RS-485 troubleshooting

3.2.1 Introduction

RS-485 is the most common asynchronous voltage standard in use today for multi-drop communication systems since it is very resistant to noise, can send data at high speeds (up to 10 Mbps), can be run for long distances (5 km at 1200 Bps, 1200m at 90 Kbps), and is easy and cheap to use.

The RS-485 line drivers/receivers are differential chips. This means that the TX and RX wires are referenced to each other. A one is transmitted, for example, when one of the lines is +5 Volts and the other is 0 Volts. A zero is then transmitted when the lines reverse and the line that was + 5 volts is now 0 volts and the line that was 0 Volts is now +5 volts. In working systems the voltages are usually somewhere around +/– 2 volts with reference to each other. The indeterminate voltage levels are +/– 200 mV. Up to 32 devices can be connected on one system without a repeater. Some systems allow the connection of five legs with four repeaters and get 160 devices on one system.

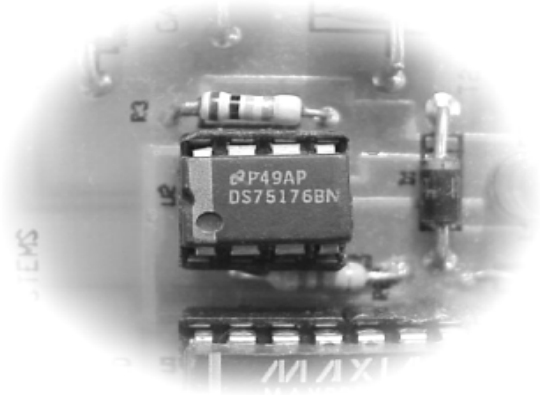


Figure 3.6
RS-485 Chip

Resistors are sometimes used on RS-485 systems to reduce noise, common mode voltages and reflections.

Bias resistors of values from 560 Ohms to 4k Ohms can sometimes be used to reduce noise. These resistors connect the B+ line to + 5 volts and the A- line to ground. Higher voltages should not be used because anything over +12 volts will cause the system to fail. Unfortunately, sometimes these resistors can increase the noise on the system by allowing a better path for noise from the ground. It is best not to use bias resistors unless required by the manufacturer.

Common mode voltage resistors usually have a value between 100k and 200k Ohms. The values will depend on the induced voltages on the lines. They should be equal and as high as possible and placed on both lines and connected to ground. The common mode voltages should be kept less then +7 Volts, measured from each line to ground. Again, sometimes these resistors can increase the noise on the system by allowing a better path for noise from the ground. It is best not to use common mode resistors unless required by the manufacturer or as needed.

The termination resistor value depends on the cable used and is typically 120 Ohms. Values less than 110 Ohms should not be used since the driver chips are designed to drive

a load resistance not less than 54 Ohms, being the value of the two termination resistors in parallel plus any other stray resistance in parallel. These resistors are placed between the lines (at the two furthest ends, not on the stubs) and reduce reflections. If the lines are less than 100 meters long and speeds are 9600 baud or less, the termination resistor usually becomes redundant, but having said that, you should always follow the manufacturers' recommendations.

3.3 RS-485 vs. RS-422

In practice, RS-485 and RS-422 are very similar to each other and manufacturers often use the same chips for both. The main working difference is that RS 485 is used for 2 wire multi-drop half duplex systems and RS-422 is for 4-wire point-to-point full duplex systems. Manufacturers often use a chip like the 75154 with two RS-485 drivers on board as an RS-422 driver. One driver is used as a transmitter and the other is dedicated as a receiver. Because the RS-485 chips have three states, TX, RX and high impedance, the driver that is used as a transmitter can be set to high impedance mode when the driver is not transmitting data. This is often done using the RTS line from the RS-232 port. When the RTS goes high (+ voltage) the transmitter is effectively turned off by being put the transmitter in the high impedance mode. The receiver is left on all the time, so data can be received when it comes in. This method can reduce noise on the line by having a minimum of devices on the line at a time.

3.4 RS-485 installation

Installation rules for RS-485 vary per manufacturer and since there are no standard connectors for RS-485 systems, it is difficult to define a standard installation procedure. Even so, most manufacture procedures are similar. The most common type of connector used on most RS-485 systems is either a one-part or two-part screw connector. The preferred connector is the 2-part screw connector with the sliding box under the screw (phoenix type). Other connectors use a screw on top of a folding tab. Manufacturers sometimes use the DB-9 connector instead of a screw connector to save money. Unfortunately, the DB-9 connector has problems when used for multidrop connections. The problem is that the DB-9 connectors are designed so that only one wire can be inserted per pin. RS-485 multidrop systems require the connection of two wires so that the wire can continue down the line to the next device. This is a simple matter with screw connectors, but it is not so easy with a DB-9 connector. With a screw connector, the two wires are twisted together and inserted in the connector under the screw. The screw is then tightened down and the connection is made. With the DB-9 connector, the two wires must be soldered together with a third wire. The third wire is then soldered to the single pin on the connector.

Note: When using screw connectors, the wires should NOT be soldered together. Either the wires should be just twisted together or a special crimp ferrule should be used to connect the wires before they are inserted in the screw connector.



Figure 3.7
A bad RS-485 connection

Serious problems with RS-485 systems are rare (that is one reason it is used) but having said that, there are some possible problems that can arise in the installation process:

- The wires get reversed. (e.g. black to white and white to black)
- Loose or bad connections due to improper installation
- Excessive electrical or electronic noise in the environment
- Common mode voltage problems
- Reflection of the signal due to missing or incorrect terminators
- Shield not grounded, grounded incorrectly or not connected at each drop
- Starring or tee-ing of devices (i.e. long stubs)

To make sure the wires are not reversed, check that the same color is connected to the same pin on all connectors. Check the manufacturer's manual for proper wire color codes.

Verifying that the installers are informed of the proper installation procedures can reduce loose connections. If the installers are provided with adjustable torque screwdrivers, then the chances of loose or over tightened screw connections can be minimized.

3.5 Noise problems

RS-485, being a differential type of circuit, is resistant to receiving common mode noise. There are five ways that noise can be induced into an RS-485 circuit.

- Induced noise on the A/B lines
- Common mode voltage problems
- Reflections
- Unbalancing the line
- Incorrect shielding

3.5.1 Induced noise

Noise from the outside can cause communication on an RS-485 system to fail. Although the voltages on an RS-485 system are small (+/– 5 volts), because the output of the receiver is the difference of the two lines, the voltage induced on the two lines must be different. This makes RS-485 very tolerant to noise. The communications will also fail if the voltage level of the noise on the either or both lines is outside of the minimum or maximum RS-485 specification. Noise can be detected by comparing the data communication being transmitted out of one end with the received communication at the other (assuming no broken wire.) The protocol analyzer is plugged into the transmitter at one end and the data monitored. If the data is correct, the protocol analyzer is then plugged into the other end and the received data monitored. If the data is corrupt at the received end, then the noise on that wire may be the problem. If it is determined that the noise problem is caused by induced noise on the A or B lines it may be best to move the RS-485 line or the offending noise source away from each other.

Excessive noise is often due to the close proximity of power cables. Another possible noise problem could be caused by an incorrectly installed grounding system for the cable shield. Installation standards should be followed when the RS-485 pairs are installed close to other wires and cables. Some manufacturers suggest biasing resisters to limit noise on the line while others dissuade the use of bias resistors completely. Again, the procedure is to follow the manufacturer's recommendations. Having said that, it is usually found that biasing resisters are of minimal value, and that there are much better methods of reducing noise in an RS-485 system.

3.5.2 Common mode noise

Common mode noise problems are usually caused by a changing ground level. The ground level can change when a high current device is turned on or off. This large current draw causes the ground level as referenced to the A and B lines to rise or decrease. If the voltages on the A or B line are raised or lowered outside of the minimum or maximum as defined by the manufacturer specifications, it can prohibit the line receiver from operating correctly. This can cause a device to float in and out of service. Often, if the common mode voltage gets high enough, it can cause the module or device to be damaged. This voltage can be measured using a differential measurement device like a handheld digital voltmeter. The voltage between A and ground and then B to ground is measured. If the voltage is outside of specifications then resistors of values between 100K ohm and 200K ohm are placed between A and ground and B and ground. It is best to start with the larger value resistor and then verify the common mode voltage. If it is still too high, try a lower resistor value and recheck the voltage. At idle the voltage on the A line should be close to 0 and the B line should be between 2 and 6 volts. It is not uncommon for an RS-485 manufacturer to specify a maximum common voltage value of +12 and –7 volts, but it is best to have a system that is not near these levels. It is important to follow the manufacturer's recommendations for the common mode voltage resistor value or whether they are needed at all.

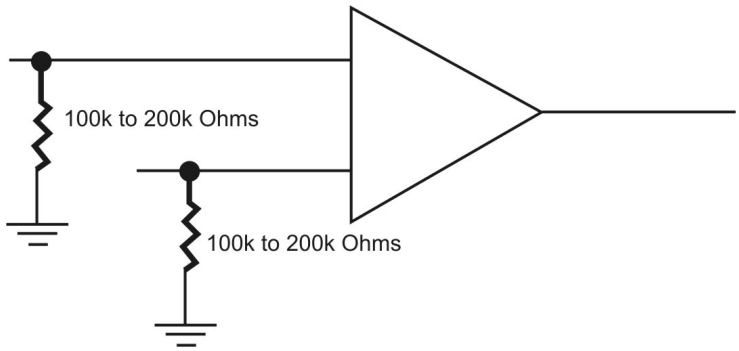


Figure 3.8
Common mode resistors

Note: When using bias resistors, neither the A nor the B line on the RS-485 system should ever be raised higher than +12 volts or lower than - 7 volts. Most RS-485 driver chips will fail if this happens. It is important to follow the manufacturer recommendations for bias resistor values or whether they are needed at all.

3.5.3 Reflections or ringing

Reflections are caused by the signal reflecting off the end of the wire and corrupting the signal. It usually affects the devices near the end of the line. It can be detected by placing a balanced ungrounded oscilloscope across the A and B lines. The signal will show ringing superimposed on the square wave. A termination resistor of typically 120 Ohms is placed at each end of the line to reduce reflections. This is more important at higher speeds and longer distances.

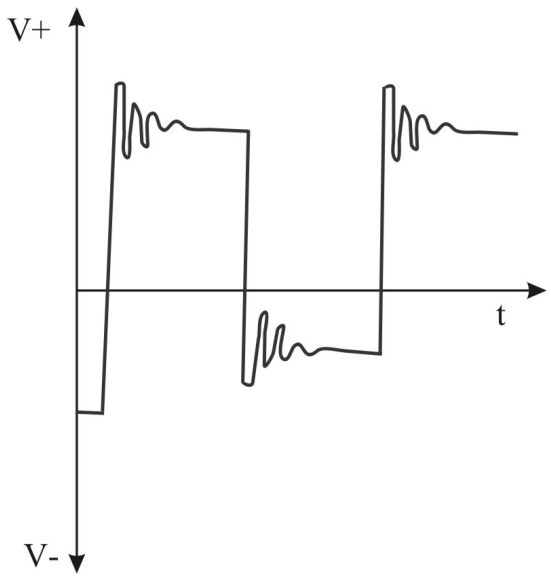


Figure 3.9
Ringing on an RS-485 signal

3.5.4 Unbalancing the line

Unbalancing the line does not actually induce noise, but it does make the lines more susceptible to noise. A line that is balanced will more or less have the same resistance,

capacitance and inductance on both conductors. If this balance is disrupted, the lines then become affected by noise more easily. There are a few ways most RS-485 lines become unbalanced:

- Using a star topology
- Using a 'tee' topology
- Using unbalanced cable
- Damaged transmitter or receiver

There should, ideally, be no stars or tees in the RS-485-bus system. If another device is to be added in the middle, a two-pair cable should be run out and back from the device. The typical RS-485 system would have a topology that would look something like the following:

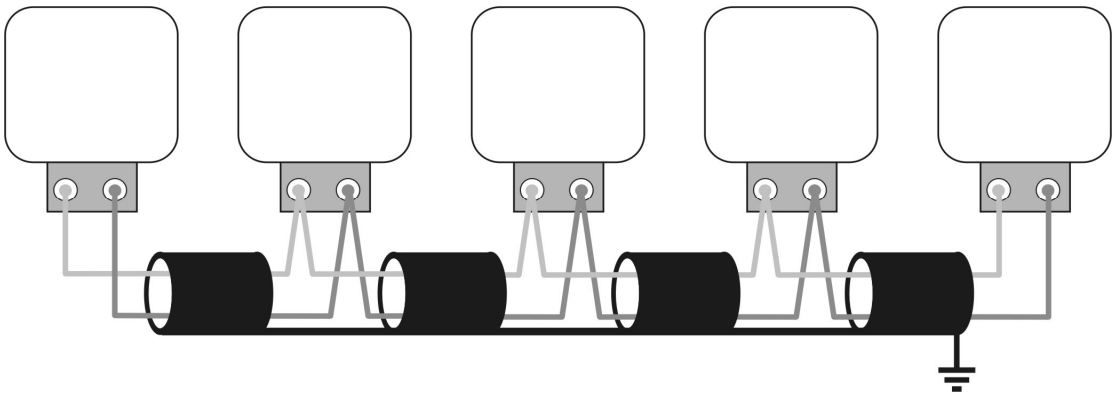


Figure 3.10
A typical RS-485

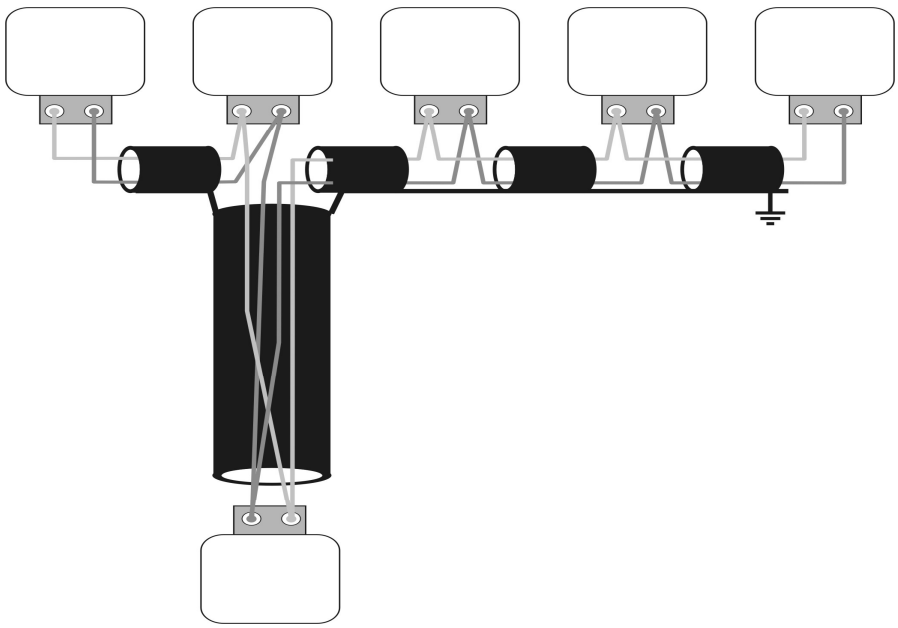


Figure 3.11
Adding a new device to a RS-485 bus

The distance between the end of the shield and the connection in the device should be no more than 10 mm or 1/2 inch. The end of the wires should be stripped only far enough to fit all the way into the connector, with no exposed wire outside the connector. The wire

should be twisted tightly before insertion into the screw connector. Often, installers will strip the shield from the wire and connect the shields together at the bottom of the cabinet. This is incorrect, as there would be from one to two meters of exposed cable from the terminal block at the bottom of the cabinet to the device at the top. This exposed cable will invariably receive noise from other devices in the cabinet. The pair of wires should be brought right up to the device and stripped as mentioned above.

3.5.5 Shielding

The choices of shielding for an RS-485 installation are:

- Braided
- Foil (with drain wire)
- Armored

From a practical point of view, the noise reduction difference between the first two is minimal. Both the braided and the foil will provide the same level of protection against capacitive noise. The third choice, armored cable has the distinction of protecting against magnetic induced noise. Armored cable is much more expensive than the first two and therefore braided and the foil types of cable are more popular. For most installers, it is a matter of personal choice when deciding to use either braided or foil shielded wire.

With the braided shield, it is possible to pick the A and B wires between the braids of the shield without breaking the shield. If this method is not used, then the shields of the two wires should be soldered or crimped together. A separate wire should be run from the shield at the device down to the ground strip in the bottom of the cabinet, but only one per bus, not per cabinet. It is incorrect in most cases to connect the shield to ground in each cabinet, especially if there are long distances between cabinets.

3.6 Test equipment

When testing or troubleshooting an RS-485 system, it is important to use the right test equipment. Unfortunately, there is very little in generic test equipment specifically designed for RS-485 testing. The most commonly used are the multimeter, oscilloscope and the protocol analyzer. It is important to remember that both of these types of test equipment must have floating differential inputs. The standard oscilloscope or multimeter each has their specific uses in troubleshooting an RS-485 system.

3.6.1 Multimeter

The multimeter has three basic functions in troubleshooting or testing an RS-485 system.

- Continuity verification
- Idle voltage measurement
- Common mode voltage measurement

Continuity verification

The multimeter can be used before start-up to check that the lines are not shorted or open. This is done as follows:

- Verify that the power is off
- Verify that the cable is disconnected from the equipment
- Verify that the cable is connected for the complete distance
- Place the multimeter in the continuity check mode

- Measure the continuity between the A and B lines.
- Verify that it is open.
- Short the A and B at the end of the line.
- Verify that the lines are now shorted.
- Un-short the lines when satisfied that the lines are correct.

If the lines are internally shorted before they are manually shorted as above, then check to see if an A line is connected to a B line. In most installations the A line is kept as one color wire and the B is kept as another. This procedure keeps the wires away from accidentally being crossed.

The multimeter is also used to measure the idle and common mode voltages between the lines.

Idle voltage measurement

At idle the master usually puts out a logical "1" and this can be read at any station in the system. It is read between A and B lines and is usually somewhere between –1.5 volts and –5 volts (A with respect to B). If a positive voltage is measured, it is possible that the leads on the multimeter need to be reversed. The procedure for measuring the idle voltage is as follows:

- Verify that the power is on
- Verify that all stations are connected
- Verify that the master is not polling
- Measure the voltage difference between the A– and B+ lines starting at the master
- Verify and record the idle voltage at each station

If the voltage is zero, then disconnect the master from the system and check the output of the master alone. If there is idle voltage at the master, then plug in each station one at a time until the voltage drops to or near zero. The last station probably has a problem.

Common mode voltage measurement

Common mode voltage is measured at each station, including the master. It is measured from each of the A and B lines to ground. The purpose of the measurement is to check if the common mode voltage is getting close to maximum tolerance. It is important therefore to know what the maximum common mode voltage is for the system. In most cases, it is +12 and –7 volts. A procedure for measuring the common mode voltage is:

- Verify that the system is powered up.
- Measure and record the voltage between the A and ground and the B and ground at each station.
- Verify that voltages are within the specified limits as set by the manufacturer.

If the voltages are near or out of tolerance, then either contact the manufacturer or install resistors between each line to ground at the station that has the problem. It is usually best to start with a high value such as 200k Ohms 1/4 watt and then go lower as needed. Both resistors should be of the same value.

3.6.2 Oscilloscope

Oscilloscopes are used for:

- Noise identification
- Ringing
- Data transfer

Noise identification

Although the Oscilloscope is not the best device for noise measurement, it is good for detection of some types of noise. The reason the oscilloscope is not that good at noise detection is that it is a two dimensional voltmeter; whereas the effect of the noise is seen in the ratio of the power of a signal vs. the power of the noise. But having said that, the oscilloscope is useful for determining noise that is constant in frequency. This can be a signal such as 50/60 Hz hum, motor induced noise or relays clicking on and off. The oscilloscope will not show intermittent noise, high frequency radio waves or the power ratio of the noise vs. the signal.

Ringing

Ringing is caused by the reflection of signals at the end of the wires. It happens more often on higher baud rate signals and longer lines. The oscilloscope will show this ringing as a distorted square wave.

As mentioned before, the 'fix' for ringing is a termination resistor at each end of the line. Testing the line for ringing can be done as follows:

- Use a two-channel oscilloscope in differential (A-B) mode
- Connect the probes of the oscilloscope to the A and B lines. Do NOT use a single channel oscilloscope, connecting the ground clip to one of the wires will short that wire to ground and prevent the system from operating
- Setup the oscilloscope for a vertical level of around 2 volts per division
- Setup the oscilloscope for horizontal level that will show one square wave of the signal per division
- Use an RS-485 driver chip with a TTL signal generator at the appropriate baud rate. Data can be generated by allowing the master to poll, but because of the intermittent nature of the signal, the oscilloscope will not be able to trigger. In this case a storage oscilloscope will be useful.
- Check to see if the waveform is distorted

Data transfer

Another use for the oscilloscope is to verify that data is being transferred. This is done using the same method as described for observing ringing, and by getting the master to send data to a slave device. The only difference is the adjustment of the horizontal level. It is adjusted so that the screen shows complete packets. Although this is interesting, it is of limited value unless noise is noted or some other aberration is displayed.

3.6.3 Protocol analyzer

The protocol analyzer is a very useful tool for checking the actual packet information. Protocol analyzers come in two varieties, hardware and software. Hardware protocol analyzers are very versatile and can monitor, log and interpret many types of protocols.

When the analyzer is hooked-up to the RS-485 system, many problems can be displayed such as:

- Wrong baud rates
- Bad data
- The effects of noise
- Incorrect timing
- Protocol problems

The main problem with the hardware protocol analyzer is the cost and the relatively rare use of it. The devices can cost from US$ 5000 to US$ 10000 and are often used only once or twice a year.

The software protocol analyzer, on the other hand, is cheap and has most of the features of the hardware type. It is a program that sits on the normal PC and logs data being transmitted down the serial link. Because it uses existing hardware, (the PC) it can be a much cheaper but useful tool. The software protocol analyzer can see and log most of the same problems a hardware type can.

The following procedure can be used to analyze the data stream:

- Verify that the system is on and the master is polling.
- Set up the protocol analyzer for the correct baud rate and other system parameters.
- Connect the protocol analyzer in parallel with the communication bus.
- Log the data and analyze the problem.

3.7 Summary

Installation

- Are the connections correctly made?
- What is the speed of the communications?
- What is the distance between the equipment?
- Is it a noisy environment?
- Is the software setup correctly?
- Are there any tees or stars in the bus?

Troubleshooting new and old systems

- Verify that there is power to the equipment
- Verify that the connectors are not loose
- Verify that the wires are correctly connected
- Check that a part, board or module has not visibly failed

Mechanical problems on new systems

- Keep the wires short, if possible
- Stranded wire should be used instead of solid wire (stranded wire will flex.)
- Only one wire should be soldered in each pin of the connector
- Bare wire should not be showing out of the pin of the connector
- The back shell should reliably and properly secure the wire

Setup problems on new systems

- Is the software communications setup at both ends for 8N1, 7E1 or 7O1?

- Is the baud rate the same at both devices? (1200,4800,9600,19200 etc.)
- Is the software setup at both ends for binary, hex or ASCII data transfer?
- Is the software setup for the proper type of control?

Noise problems on new systems

- Induced noise on the A or B lines?
- Common mode voltage noise?
- Reflection or ringing?